How Does Poverty Affect Student Achievement

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Introduction

As of the last 5 years, education inequality has been a central topic in economics and education research. Even though access to education has been increasing throughout the years, student achievement gaps remain large between students from different socioeconomic backgrounds. Achievement gaps not only show up when students take standardized tests, but also predict graduation rates, college-going rates, and future earnings. Poverty is among many factors that cause these gaps and is one of the most prevalent and difficult to tackle.

Students from poor families are disadvantaged in many ways before they even enter school. Not only do they have less educational resources at their disposal, such as books or computers, but they also must grapple with challenges of having an unstable place to live, not knowing where their next meal is coming from, and higher levels of stress. These things take a toll on cognitive development and academic performance. For students who grow up in poor households, the struggles don't stop even after they make it into school. Schools in low-income areas face a host of challenges that make it harder for them to provide students with the level of education that higher-income districts can.

The reason why this topic is of particular importance for economists is because education is one of the key drivers of human capital. Economic theory says that workers are productive because they are skilled; and they are skilled because they invested in their education and training. This implies that if children living in poverty are unable to receive a quality education, this will decrease the overall skill level in the economy and stunt economic growth. Education also affects income inequality, because if students from lower-income families receive a lower-quality education, they will not be able to reach their full potential and will remain in poverty.

We found this topic interesting because while the idea that poverty affects education is frequently talked about, there isn't much analysis of this relationship on a detailed analytical level. We were expecting to find that poverty had a negative impact on student achievement. What we were unsure of was if this was true for *all* districts or only for districts with higher poverty rates.

The purpose of this paper is to analyze the effect of poverty on student achievement and how this relationship varies across school districts in America. More specifically, the paper looks at if the relationship between poverty and achievement is linear or nonlinear. The hypothesis of this paper is that poverty has a negative impact on student achievement. Alternatively, this paper examines if the marginal effect of poverty on achievement is constant. Perhaps districts that have recently become poor see larger drops in performance than districts that have been poor for many years.

To test this hypothesis, we combine district-level data from Stanford Education Data Archive (SEDA) with poverty estimates from American Census Bureau's Small Area Income and Poverty Estimates program (SAIPE.) The data reveals that there is indeed a strong negative correlation between poverty and student achievement. However, the relationship is indeed nonlinear. The worst impacts of poverty are seen when districts reach poverty thresholds of low to moderate levels of poverty. As districts become increasingly poor, the negative effects of each one percent increase in poverty rate lessens.

Literature Review

One hypothesis is that poverty creates school resource gaps between districts. Adam Gamoran and Bianca P. An (2016) confirm that students in high-poverty districts are more likely to go to schools with fewer resources. Schools in low-opportunity communities may experience less funding, have less-experienced teachers, and lack authentic learning opportunities. All of these features may drive differences in achievement between districts. Students may face constraints from their home environments that make it more difficult for them to do well in school. Low-income families are less likely to have access to books, tutors, and educational technology. Financial insecurity can take a psychological toll on both students and their parents. These constraints may make it more difficult for students to succeed in school.

Student achievement may be driven by families' long-term socioeconomic status. School poverty is often measured differently from general socioeconomic status (SES) measures. While SES measures often use variables that speak to long-term characteristics of the home such as parent education and occupation, school poverty measures often rely on eligibility for free or reduced-price lunch. Andrew Hegedus (2018) demonstrates that short-term measures of poverty like free lunch are highly correlated with achievement, accounting for a large portion of the achievement gap between schools. This correlation is much weaker for student growth, which suggests schools can be effective even while serving high-poverty populations. David Sparks and Christine Mulhern (2026) study a policy that counters the effects of poverty on achievement. They find that school-based counseling services improve students' educational outcomes. Their work suggests that policymakers can invest in initiatives to directly reduce poverty or help students overcome the constraints they face.

The research discussed above was made possible by newly available data on thousands of school districts. SEDA was one of the first to leverage these datasets to understand the role of socioeconomic status in educational outcomes. Few studies have estimated whether the relationship between poverty and student test scores is nonlinear. Prior research provides some reason to believe this relationship is linear; families may adapt to life in high-poverty neighborhoods, or high-poverty districts may receive more targeted policy attention. Nonetheless, we allow the relationship to be nonlinear in our empirical strategy.

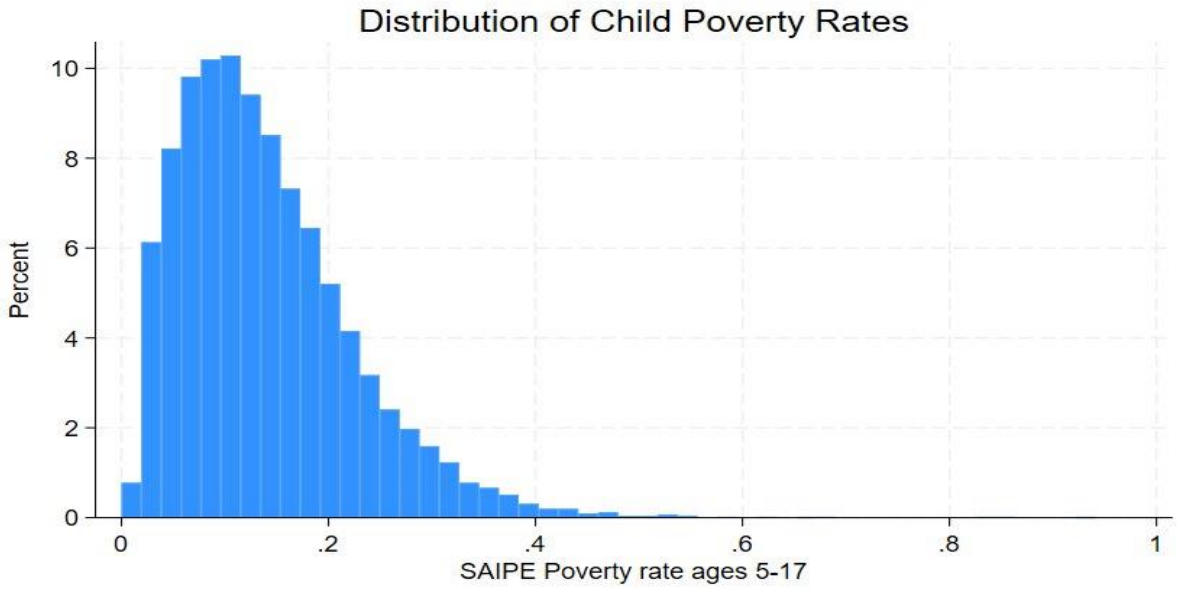
Data and Methodology

We used district-level data created by combining our sources to conduct our analysis. We collected it from SEDA's website, and we used it to obtain students' standardized achievement test scores. Standardized exams are normally administered by school districts or States to grade schoolers on their knowledge in certain subjects. Scores can then be compared from district to district or State to State. Comparisons like this are made possible by how scores are normed. Using this data, we measure student achievement by the average district score. Poverty is measured using American Community Survey data from the United States SAIPE dataset. Our explanatory variable of interest is measured by calculating the percentage of children between the ages of five and seventeen living under the poverty line in each district. Other than poverty being a plaguing factor that students may face, it was chosen because it can be used to measure a students' current economic situation.

Control variables were chosen based on how they could affect achievement and correlate poverty. The most crucial control we use is socioeconomic status average, shortened SESAVGALL. While poverty data can show current factors that affect students, SES can help

show structural long-term factors that students could be faced with. Structural factors being the status of their household over a period of time considering the income and education of parents. Variables like these let us compare the short-term effect of poverty on students while controlling for the student's long-term income status. Total enrollment was used as a proxy for school district size. School size can play a factor because larger districts may be able to provide more students with less resources than smaller districts. Additionally, we use dummy variables for district types. Our categories for district type are suburban, town, and rural. We use urban as our baseline or reference category. These variables help control for population density and other location-based factors.

After we cleaned our data, we noticed that our variable of interest, poverty, was extremely skewed to the right. As you can see in the histogram following this paragraph, most districts fell under a low poverty percentage while as rate increased the number of districts decreased. If we were to run a regression with this variable as is, we would be assuming that the relationship between poverty and achievement is linear. Our initial model without the quadratic variable made the impact not as accurate and not as significant as we hoped according to its p-value. We decided to account for this by adding a squared poverty term. This allows us to test any kind of relationship between poverty and achievement.



SAIPE Poverty Frequency Rates Among Children 5-17 (2024)

The regression model was clustered by state using the "absorb(fips)" command in Stata. In other words, we are accounting for any differences that may exist between states. Such as, education laws, funding, and even differences in testing. Our final dataset contains around 97,000 observations.

Preliminary Descriptive Statistics

Before running the regression model, looking at some basic patterns that emerge in the raw data, The first thing that jumps out when looking at this dataset is the distribution of poverty rates across districts. As said previously, poverty rates appear to be heavily right-skewed. That is, districts exhibit relatively low poverty rates with a few districts exhibiting abnormally high rates of poverty. To go more in-depth on this topic, if poverty rates are included in a linear regression as a predictor of test scores, the districts with extremely high poverty rates will drag the line of best fit down. In other words, these districts will have disproportionate influence over the slope

calculated for poverty. Because most districts have poverty rates that aren't astronomically high, this could cause us to overestimate the impact of poverty on test scores.

Another thing to notice when looking at poverty and scores is that they are moderate to strongly correlated negatively. As poverty goes up, test scores go down. But this relationship isn't perfectly linear as stated previously. Instead, it looks like when districts are at lower levels of poverty, a small increase in poverty will lead to a larger decrease in test scores. But at higher levels of poverty, when poverty increases by a small amount, test scores don't decrease as much. This would make sense if we think about the marginal effect of poverty on test scores. If a district that was not used to having many poor students suddenly started experiencing very high poverty levels, we might expect a student's performance to decrease significantly. However, if a district consistently ranks high in terms of poverty, then an increase in poverty will not affect students' performance as much. This observation supports our decision to have a nonlinear specification for poverty. The figure below gives a visual explanation to how our variables correlate to each other.



Correlation Heat Map Among all Key Variables

The final pattern we wanted to note before we run the regression is the positive association between SES and test scores. As the level of SES in a district increases, so do test scores. This makes sense if we think about how students who grow up in poor households might still perform well on tests if they come from districts or households with high SES. In other words, students in poorer districts might have some structural advantages that help them despite high levels of poverty. On the other hand, if a district has been at high levels of poverty for a long time, high SES families may start to leave the district, causing overall SES to fall, and may need additional assistance from policy makers to help support these districts' performance.

Further Discussion of Descriptive Statistics

Before running a regression, we always want to become familiar with the data we are using. While descriptive statistics such as means and standard deviations are useful for quickly understanding the nature of our variables, it can be beneficial to further interpret these values. To start, consider the poverty measure, which is measured as the percentage of children between 5-17 that are living in households below the poverty line. As we can see from the minimum and maximum values, there are districts that have near 0 levels of poverty and those that approach 60% poverty levels. The average district has around 13% of students living in poverty.

However, because the median is much smaller than the mean, we know that this data is right skewed. There must be many districts with low poverty rates and fewer districts with abnormally high poverty rates that drive the mean up. This pattern can be important when running regressions. Recall from the example paper that highly skewed variables can affect our regression. In this case, the high skewness of poverty prompted us to include a squared term of poverty. Moving onto student achievement, we see that this also varies by district. There are quite a few districts that perform well on standardized tests, and many districts that score far below average. On the other hand, there seems to be less dispersion in districts that score well on tests than those that don't. This could mean that while districts that score well have similar characteristics that help their students achieve, underperforming districts vary more. Like poverty, SESAVGALL also varies by district.

Since the mean is roughly halfway between the minimum and maximum value, it seems that there are a decent amount of districts that fall in the middle range of the SES measure. This will be important to keep in mind since we expect SES to be highly correlated with both poverty

and student achievement. Lastly, we can see that total enrollment, or district size, ranges from very small districts to some massive districts. Without that districts on the far ends of the enrollment spectrum could be outliers. For that reason, we decided to control for enrollment in our regression. We can see from the correlation table that poverty is indeed negative correlated with test scores, which we would expect from theory. Similarly, SES is positively correlated with test scores, meaning that students who have come from high SES backgrounds will likely have higher test scores.

Regression Results

	(1) Administrative District Subject-Grade-Year Average, All Students, CS	(2) Administrative District Subject-Grade-Year Average, All Students, CS	(3) Administrative District Subject-Grade-Year Average, All Students, CS	(4) Administrative District Subject-Grade-Year Average, All Students, CS	(5) Administrative District Subject-Grade-Year Average, All Students, CS
SAIPE POV.	-3.165*** (0.015)	-0.128*** (0.028)	-0.122*** (0.028)	-0.103*** (0.028)	-2.302*** (0.278)
SESAVG		0.324*** (0.003)	0.324*** (0.003)	0.324*** (0.003)	0.294*** (0.021)
total enrollment			-0.000*** (0.000)	-0.000*** (0.000)	-0.000* (0.000)
suburban locale				0.085*** (0.005)	0.011 (0.014)
town locale				0.096*** (0.005)	0.057* (0.026)
rural locale				0.076*** (0.005)	0.020 (0.020)
poverty_sq					3.908*** (0.718)
Observations	98,331	97,977	97,977	97,977	97,977

Determinants of SEDA Test Scores 2024
Standard errors in parentheses
Note: Model 3 includes State Fixed Effects.
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The regression table tells us that there is a negative relationship between poverty and student achievement. For all model specifications, the coefficient on poverty is negative and significant. The coefficient on poverty is quite large in the baseline specification where only poverty is included as an independent variable. This would lead us to believe that poverty has a meaningful impact on student achievement. However, once we control other variables in the model, the coefficient gets smaller (but remains significant). This would suggest that our poverty variable is picking up some of the same variation that is captured by the socioeconomic status and district control variables.

The regression table also includes the squared poverty term. The coefficient on poverty is negative and significant while the coefficient on poverty squared is positive and significant. These signs are exactly what we would expect if the marginal effect of poverty on test scores decreases as the poverty level increases. The first few increments of poverty have the biggest negative effect on test scores. As poverty increases, the negative effect that each additional percentage point has on test scores gets smaller. In econometric terms, we say that the marginal effect of poverty on test scores is subject to diminishing marginal effects.

Socioeconomic status is by far one of the strongest variables in terms of its relationship to test scores. The coefficient on SES is positive and extremely significant across all model specifications. This would suggest that districts with "better" long-run socioeconomic conditions will have higher test scores on average. Total Enrollment has a small negative effect on test scores. This relationship makes sense, but the overall effect is small. Total enrollment may capture some of the small benefits of economies of scale districts lose as they become too large. Finally, the geography variables are interesting. Suburban and town are both associated with higher test scores than urban districts while rural is somewhat mixed.

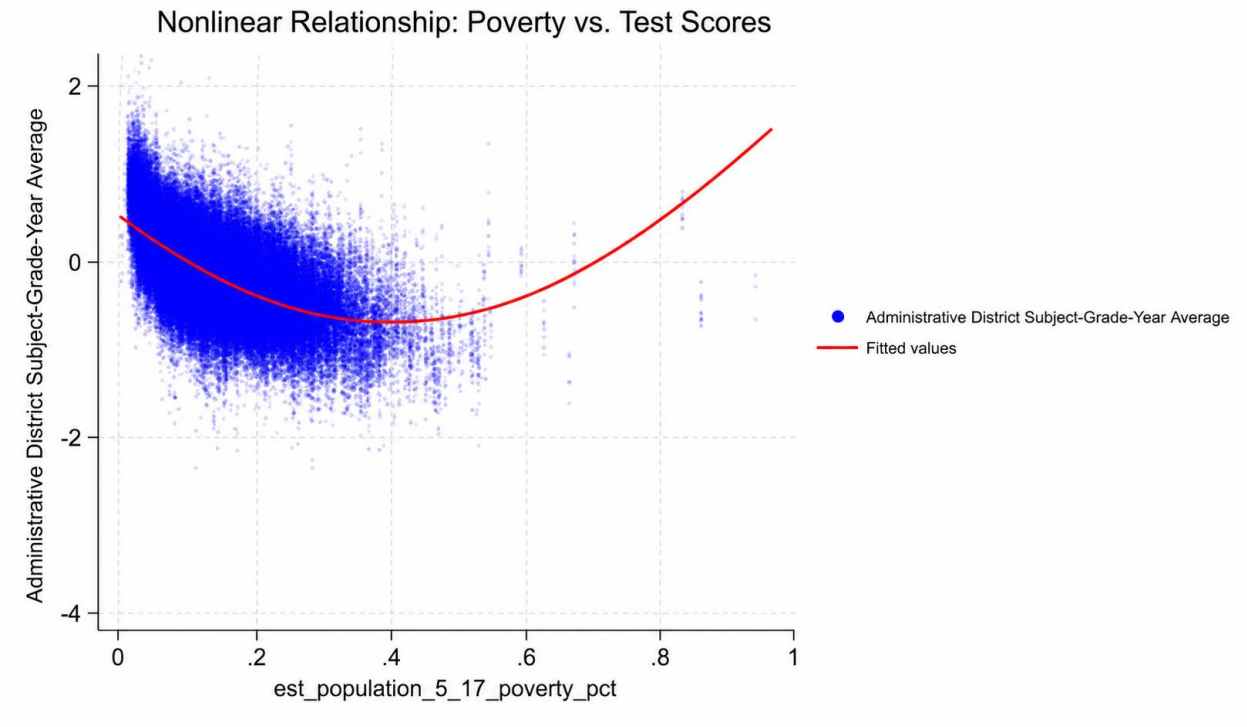
$$Y_{ids} = \beta_0 + \beta_1 \text{Poverty}_{ds} + \beta_2 \text{Poverty}_{ds}^2 + \beta_3 \text{SES}_{ds} + \beta_4 \text{Enrl}_{ds} + \mathbf{L}_{ds}\Gamma + \alpha_s + \epsilon_{ids}$$

Interpretation of Regression Results

In order for regression results to be useful, we need to interpret each individual coefficient meaningfully. We already know that the coefficient on poverty will be negative and significant. This relationship should make sense and agrees with the literature on school testing. Let's go one step further though and put this regression coefficient into terms we can use. A one percentage point increase in poverty can be expected to decrease test scores by a certain number of points. Let's call it "x". In other words, an increase in poverty from 20% to 30% can be expected to decrease district average test scores by (x*10) points, which is a pretty sizable effect.

Remember, we also added a poverty squared variable to the regression. But the coefficient on poverty is still negative and significant, while the coefficient on poverty squared is positive and significant. These two signs imply that the marginal effect of poverty on test scores is negative, but decreases as the poverty rate increases. Economists love to use the phrase "diminishing marginal effects", which basically means what we described above. Small increases in poverty will have a large impact when the district poverty level is low. As the poverty level gets very high, continued increases in poverty will have a smaller negative impact on test scores. Graphically, the marginal effect looks like this:

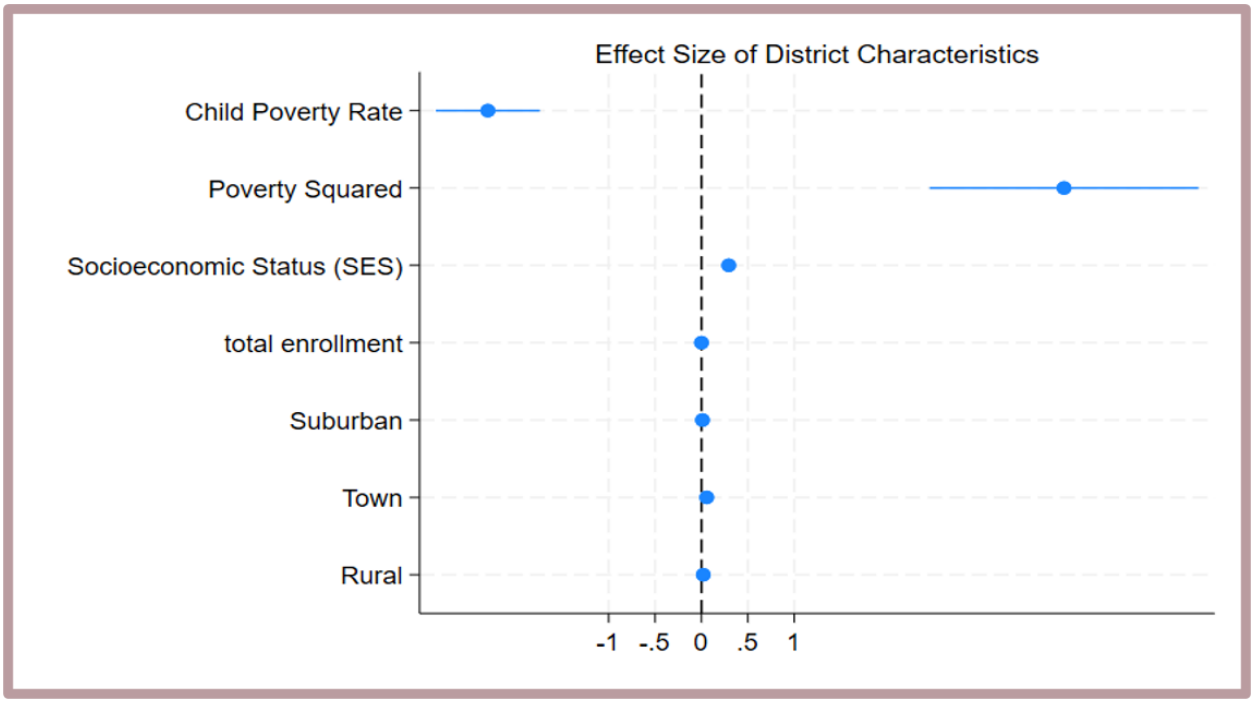
Scatter Plot



Let's think about why this pattern would exist. If a district free of poverty suddenly increases to 1%, it will experience new test taking challenges that are associated with poverty, but it will go from having almost no disadvantage to being considered poor. As the district gets poorer, (say 60%) increases in poverty will still hurt test scores, but the children may already be experiencing many of the drawbacks associated with living in poverty. SES, as expected, has a large positive coefficient. This relationship should send a clear message that long-term structural variables matter when it comes to school testing. Regardless of a district's current poverty level, if they have a high SES students will achieve at higher levels than students with lower SES.

Enrollment has a small negative effect on test scores. While not exactly meaningful on its own, this effect tells us that as districts get larger, they tend to see lower test scores. This could

be due to any number of factors we talk about above. Our geo-political variables were also valuable. Being in a suburban district is objectively much better than being in an urban district. This may be due to school funding, population density differences, or any number of other factors. Town districts also outperformed urban districts while rural districts showed mixed results.



Key Points:

- Socioeconomic Status is the largest positive coefficient.
- Poverty has a large effect on test scores
- Poverty Squared is positive which showcases how test scores begin to flatten out as poverty rates increase or "go deeper."
- Students tend to outperform their urban baseline counterpart.

Longer discussion of Economic interpretation

Viewing these results as an Economist, poverty influences student achievement through limited formation of human capital due to lack of opportunities for students in high-poverty districts. Furthermore, poverty impact levels are nonlinear, and this helps us better understand this dynamic. Seeing that poverty has a decreasing marginal effect tells us that jumping from 0% poverty to some small amount has larger negative consequences than increasing from a higher level of poverty. We think this could mean that there is some sort of shock that occurs to districts as they start becoming poorer. Perhaps, schools experience funding cuts, or there is some rush of families below the poverty line moving into the district. Either way, this would cause a large negative impact on student achievement.

However, as districts start becoming poorer over time, they may be able to counteract this effect to some degree. Maybe the district learns to cope with their new level of poverty, or they receive more funding from the state government to help programs for low-income students. While this will not inherently by itself increase test scores, it could prevent them from falling at the high rate that we see districts with lower poverty levels. We also think that the large positive impact that SES has on student achievement could be evidence for the importance of these types of long-term factors. If districts with higher poverty levels are still able to trump the effects of poverty through high SES, there could likely be a cumulative effect of long-term programs that help students from low-income families.

Discussion

First, we can confirm that poverty is one of the driving factors of student achievement. This helps us to confirm that economic factors are some of the most important factors that determine success in schools. The non-linear relationship that we see could mean that the best time to provide funding to a district is when they start to see their poverty levels rise (emphasis on "could"). But we may be able to ensure that all students have access to a great education by programing initiatives at the exact moment when poverty hits a district, instead of waiting for the poverty rate to become too high. The strong relationship that is shown between socioeconomic status and student achievement could be indicative of the importance of long term investments. Instead of simply managing the poverty levels in students' schooling lives, it may be more effective to help these students out after they leave school via after school programs and such.

Limitations

We were only able to identify a correlation between poverty and student achievement and cannot say for certain that poverty causes these decreases. So more so correlation rather than causation. There is likely some level of bias in the coefficients that we are unable to control for. Similarly, by using district level data, we were unable to capture variation within districts. Students in the same district could have drastically different economic backgrounds that are not being accounted for. The only thing that we could do was add a state fixed effect variable to account for how different states are affected by things such as different funding policies local or federal. Additionally, we may still be specifying our model incorrectly. While we can see that a quadratic model greatly helps us when compared to a linear model, there is still a possibility that

the true relationship is not quadratic. Finally, and most obviously, we were unable to look at how these relationships change over time due to the fact that we are only using cross-sectional data.

Results Discussion

First, we see that regardless of what other variables we control for, poverty consistently showed up as a significant variable. As we added more control variables, the coefficient on the poverty rate became smaller, but never once did it become positive or insignificant. This led us to believe that the poor performance of students in high poverty districts is most likely due to the high poverty rate and not some other factor correlated with poverty.

Along with noticing how poverty consistently showed up as a significant variable, we can see the importance of including SES in our model. Once we account for SES, the beta coefficient for the poverty rate more than halves in size. This could mean that a lot of the poverty rate's effect on student achievement is driven by long-term factors that contribute to the SES of a district. Similar to the last point, we saw that the relationship between poverty and student performance was not linear. The negative relationship between poverty and performance decreases as the poverty rate gets higher. A potential explanation for this could be that districts with poverty rates on either extremes do not suffer nor gain as greatly from changes in their poverty rate. While districts with high poverty rates are seeing underperformance in students, they could be subject to less change than districts where poverty is starting to become an issue.

Quantitative Analysis

One aspect that would be interesting to look at is using data from multiple years to see how poverty rates and student performance change over time. By using data from the same districts over a range of years, we would be able to use various methods to better identify the causal relationship between poverty and student performance. Only problem is, that would mean that we would've needed to use time-series data. While using data at the district level is nice, it would be interesting to see how these relationships play out on an individual level. It would allow us to control for various policies that each school may have. By using data from students and their family's income, we would be able to identify how income directly causes changes in performance. Lastly, it would be interesting to look at how specific policies effect student performance. By collecting data from when certain policies were passed, we would be able to identify how things like changes in school funding or support programs effect student performance.

Conclusion

In this study, we learned that student performance is significantly impacted by the poverty rate of the district they attend. Using data from 766 districts across the United States, we were able to identify the fact that, on average, students in high poverty districts perform significantly worse than their peers in lower poverty districts. We also found that this relationship is nonlinear.

Policy Recommendations

Based on these results, we would recommend that not only should there be programs to help students who are currently living in poverty, but we should also focus on long term policies that can prevent parents from falling into poverty. This would be able to prevent students from falling into poor performing schools. As mentioned previously, we also think that there should be an increase in programs when a student's district begins to reach that poverty level. If we can assist students before their district reaches high poverty levels we can maybe guarantee that every child has the opportunity to receive a quality education. If we want our economy in the United States to continue to grow we need to assist those students who are confronted with poverty. By making sure these students can receive a quality education we are improving the economy of the town or city they live in and giving them a better chance of landing a good job.

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